SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 3.1: Turning Effect of Force — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.1: Turning Effect of Force
Driving Question	DRIVING QUESTION: Why does where you push, pull, or grip something determine how easy or hard it is to make it turn — and how do engineers and farmers in Kenya use this to design better tools?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **The Jembe Handle Puzzle — Launching the Phenomenon**	Students handled jembes (and/or observed a demonstration) with handles of different lengths and gripped at different positions. They noticed that gripping farther from the blade made turning/digging noticeably easier, while gripping near the blade required much more effort to produce the same turning action. Some students also observed that a short-handled jembe caused more fatigue than a long-handled one for the same task.	At this stage students have NOT yet learned the formal scientific explanation. They have established the anchoring phenomenon and generated initial predictions: handle length and grip position both appear to affect how easy it is to make the jembe rotate. These observations are logged on the Driving Question Board (DQB) as questions to be answered across the unit.	No causal explanation is formalised in Lesson 1 — this is intentional and pedagogically important. The lesson functions as a 'need-to-know' hook. Teacher note: resist the urge to give the answer; instead, validate student predictions and record all competing ideas on the DQB. Common misconception to flag (but not resolve yet): many students will say 'a longer handle is heavier, so it does more work' — note this for targeted correction in Lesson 3.
Lesson 2 **Moments of a Force at a Point — The Metre Rule Balancing Investigation**	Students balanced a metre rule on a central pivot and hung known weights (or used standard masses) at measured distances on both sides. They systematically varied force and distance and recorded every balanced configuration in a results table. Key observation: in every balanced case, the product Force × Distance on the left equalled Force × Distance on the right ($F_1 \times d_1 = F_2 \times d_2$), even when the individual forces and distances were very different.	A force can cause a turning effect about a pivot. The turning effect depends on BOTH the size of the force AND how far it acts from the pivot. The data pattern $F_1 \times d_1 = F_2 \times d_2$ is the empirical foundation of the Principle of Moments. Students learn to distinguish clockwise and anti-clockwise turning directions and to record moments systematically.	The investigation provides concrete, student-generated evidence that turning effect is a product of two variables — force and distance — not force alone. Teacher note: emphasise that this is the same reason a farmer gripping the jembe handle farther from the blade achieves greater turning effect with the same muscular force. Cold-call students to connect each balanced configuration back to the driving question. Pedagogical caution: ensure students measure perpendicular distance from the pivot, not along the ruler at an

			angle — a common procedural error at this level.
Lesson 3 **Formalising the Moment Equation: M = F × d and the Principle of Moments**	Students reviewed their Lesson 2 data tables and, guided by teacher questioning, derived the formal equation from their own results. Additional worked examples using Kenyan contexts (door hinges, panga handles, jembe grips) were analysed to consolidate the pattern.	The moment of a force $M = F \times d$, where F is the applied force in Newtons and d is the perpendicular distance from the pivot in metres. The SI unit of moment is the Newton-metre (Nm). The Principle of Moments states: for a body in rotational equilibrium, the sum of clockwise moments equals the sum of anti-clockwise moments about any point.	Students can now provide the first formal answer to the driving question: gripping farther from the pivot (hinge or blade) increases d, which increases M even if F stays the same — making turning easier. Teacher note: explicitly return to the jembe phenomenon and ask students to annotate a diagram of a jembe with F, d, and M. This is a critical DQB update moment. Address the Lesson 1 misconception about handle weight here — it is distance, not mass of handle, that drives the larger moment.
Lesson 4 **Anti-Parallel Forces and Couples: When Two Pushes Make a Turn**	Students applied two equal forces in opposite directions to a steering-wheel model (or a rigid card with two force meters) and observed that the object rotated without translating (moving sideways). They compared this to a single force applied at one side. Demonstration contexts included tightening a water tap in Mombasa-style plumbing fittings and turning a bicycle handlebar.	A COUPLE is a pair of equal, parallel, anti-parallel forces whose lines of action do not coincide. The resultant force of a couple is zero (no translation), but the resultant moment is NOT zero — the object rotates. The moment of a couple = $F \times d$, where d is the perpendicular distance between the two lines of action (the 'couple arm').	The couple concept explains why we instinctively use both hands on a steering wheel, tap, or hand drill — applying forces on opposite sides doubles the effective couple arm compared to pushing with one hand at the edge. Teacher note: connect to the driving question by asking — 'Why do matatu drivers grip the steering wheel with both hands on opposite sides rather than pushing from one side?' This surfaces the couple concept in lived Kenyan experience. Highlight that the net force being zero means a couple causes pure rotation — an important distinction from single-force moments.
Lesson 5 **Calculating Moments of Anti-Parallel Forces — Numerical Problems**	Students worked through a structured set of numerical problems involving single moments, balanced beams, and couples. Problems were set in Kenyan contexts: a Kericho tea-picker using a hand pruner (secateurs), a Kisumu fisherman operating a boat rudder, and a construction worker in Nairobi using a spanner to tighten bolts. Students solved problems individually and then peer-checked using the Principle of Moments.	Students consolidated the calculation skills for: (1) single moment $M = F \times d$; (2) applying the Principle of Moments to find unknown forces or distances in balanced systems; (3) calculating the moment of a couple = $F \times d$ (couple arm). Students also learned to correctly identify the pivot and perpendicular distance in non-trivial diagrams where the force is not applied at 90°.	Numerical fluency confirms conceptual understanding — a student who can correctly calculate why moving a spanner grip from 10 cm to 30 cm triples the moment has internalised $M = F \times d$. Teacher note: common errors at this stage include (a) using the wrong distance (along the rod rather than perpendicular to the force line), and (b) confusing the couple arm with the radius. Use targeted questioning: 'Where exactly is the pivot? What is the shortest distance from the pivot to the force's line of action?' Allow 3–4 minutes of independent work before cold-calling — minimum wait time is critical for equity.

Lesson 6 **Walk-Around and Real-Life Application Mapping: Turning Effect of Force in Our Environment**	Students conducted a structured 'walk-around' activity — either physically around the school compound or through a curated photo gallery of Kenyan environments — cataloguing tools and structures that use turning effect: school gate hinges, water tap handles, wheelbarrows used by fundis, pliers in a workshop, a Rift Valley farmer's hand pump, bicycle brake levers, and a flagpole pulley system.	The Principle of Moments is universal — it governs the design of every lever, spanner, door, tap, brake, and crane. Students learned to classify real-world turning-effect applications by identifying: the pivot point, the effort force and its distance, the load force and its distance, and whether the system involves a single moment or a couple. They also recognised that engineers deliberately lengthen handles or increase couple arms to reduce the effort force required.	This lesson broadens the driving question from the jembe to the wider designed world. Students should now articulate: 'Engineers in Kenya — from those designing irrigation hand pumps in the Rift Valley to bicycle mechanics in Kisumu — apply M = F × d to ensure tools require the least possible effort.' Teacher note: the walk-around generates rich DQB additions. Ensure students update their cumulative model diagrams with at least two new real-life examples. Prompt higher-order thinking: 'If a Nairobi crane engineer needs to lift a heavier load without increasing the engine force, what design change does the Principle of Moments suggest?'
Lesson 7 **Stability, Centre of Gravity and the Turning Effect — Why Things Tip or Balance**	Students investigated how the position of an object's centre of gravity relative to its base of support determines whether it topples or remains stable. Practical activities included tilting loaded and unloaded cardboard boxes, comparing a tall thin water bottle with a short wide one, and examining why a heavily loaded mkokoteni (handcart) tips forward if cargo is placed too far in front of the axle.	The centre of gravity (CoG) is the single point through which the entire weight of an object appears to act. An object topples when a turning moment due to its weight (acting through the CoG) is not balanced by a reaction from the base of support. Stability increases when the CoG is lower and the base of support is wider. The turning effect of gravity about the tipping edge determines whether an object falls or returns to equilibrium.	Centre of gravity links directly to moments: the weight force (W) acting at a horizontal distance (d) from the tipping edge produces a moment W × d. If this moment is unbalanced, the object tips. Teacher note: connect explicitly to the driving question — 'A farmer loading a mkokoteni or stacking maize sacks on a cart must consider where the combined centre of gravity falls relative to the wheels; the Principle of Moments predicts whether the load will tip.' This lesson also lays the groundwork for Lesson 8's consolidation by unifying CoG, stability, and M = F × d into one coherent model. Flag that this concept underpins engineering safety standards for buildings and vehicles in Kenya.
Lesson 8 **Consolidation, Assessment and Reflection: The Final Explanation of Turning Effect of Force**	Students revisited the original jembe phenomenon from Lesson 1 and compared their initial predictions (DQB entries) with their current understanding. They completed a summative task: annotating a diagram of a jembe, a door, and a couple-based tap, labelling pivot, force, perpendicular distance, and calculated moments. A final DQB review session surfaced any remaining questions.	Students demonstrated mastery of the full conceptual arc: (1) M = F × d (unit: Nm); (2) Principle of Moments — clockwise moments = anti-clockwise moments at equilibrium; (3) a couple produces pure rotation with zero net force, moment = F × d (couple arm); (4) centre of gravity and stability are governed by the same turning-effect logic; (5) engineers and Kenyan farmers apply these principles deliberately in tool and structure design.	The full answer to the driving question: where you push, pull, or grip something determines the perpendicular distance d from the pivot. Since M = F × d, a larger d produces a greater turning moment for the same force — making rotation easier. Engineers designing jembes, spanners, water pumps, and cranes in Kenya maximise d (lengthen handles or increase couple arms) to reduce the effort needed. Teacher note: use this lesson to celebrate student growth — display Lesson 1 predictions alongside Lesson 8

			explanations to make learning visible. Ensure every student has updated their cumulative model diagram to include all four concept layers: single moment, Principle of Moments, couples, and CoG/stability.